

As more and more games started to rely on DirectX 9 features, the poor shader performance of nVidia's GeForce FX series became more obvious. nVidia then became desperate to hide the shortcomings of the GeForce FX range. The 5700 Ultra, as a late revision, ended up as the only FX series part with competitive shader performance. And eventually, nVidia lost its market leadership position to ATI.

With the release of the GeForce 7 series, nVidia has once again emerged as the market leader. Currently, it will be assisting Sony with the design of the graphics processor for the PlayStation 3.

ATI Inc

ATI began as an OEM, producing integrated graphics chips for large PC manufacturers. By 1987, it had evolved into an independent graphics card retailer, marketing the EGA Wonder and VGA Wonder graphics cards under their own "ATI" banner. Each of these families offered enhanced feature sets that surpassed IBM's own display adapters. In fact, 1991 saw the release of the Mach8, ATI's first "Windows accelerator" product.

Windows accelerators offloaded display processing tasks. And 1992 saw the release of the Mach32 chipset, an improvement over its predecessor. ATI had established a reputation for quality multimedia-capable cards popular with OEMs; by the late 1990s, consumers began to expect strong 3D performance, and 3Dfx and nVidia were delivering.

The first warning was seen in January 1999 with the All-in-Wonder 128, featuring the Rage 128 GL graphics chip. While the basic 16 MB version sold reasonably well, the 32 MB version did not, because it lacked 3D acceleration appropriate for the price.

ATI's first real 3D chip had been the 3D Rage II, which supported bilinear and trilinear filtering, z-buffer, and several Direct3D texture blend modes. But the pixel fill rate was only next to S3's VIRGE range, and the feature list was only next to the Matrox Mystique.